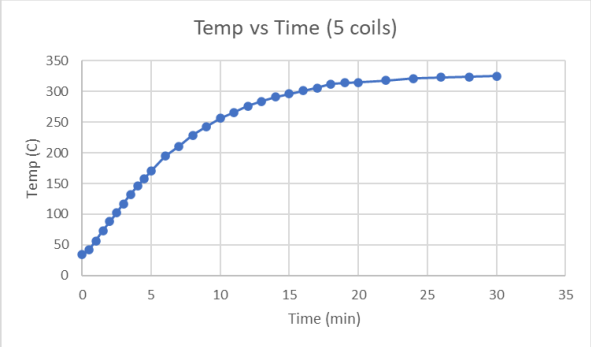
Heating Data

SS Air Test @5amps

5 coils

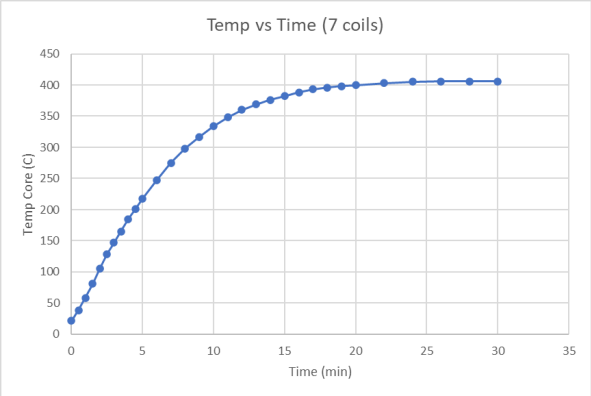
Time (min)	Temp Core (C)
0	34
0.5	42.4
1	56.5
1.5	73.5
2	87.8
2.5	102.8
3	117.2
3.5	132.3
4	145.5
4.5	157.9
5	170.7
6	194.6
7	211
8	229
9	243
10	256
11	266
12	276
13	284
14	291
15	296
16	301



17	306
18	312
19	314
20	315
22	318
24	321
26	323
28	324
30	325

7 coils

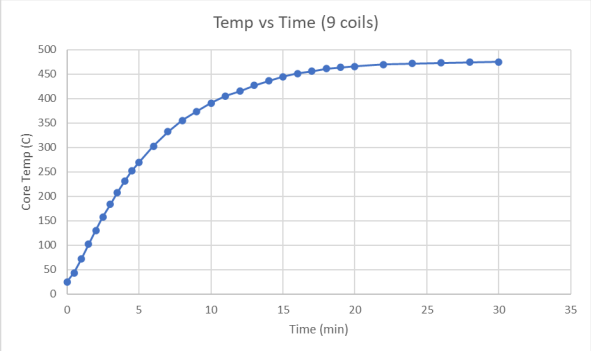
Time (min)	Temp Core (C)
0	21.8
0.5	38.8
1	58.6
1.5	81.2
2	105
2.5	128.7
3	146.8
3.5	165.1
4	184.7
4.5	201
5	218
6	247
7	275
8	298
9	317
10	334
11	348
12	360



13	369
14	376
15	382
16	388
17	393
18	396
19	398
20	400
22	403
24	405
26	406
28	406
30	406

9 coils

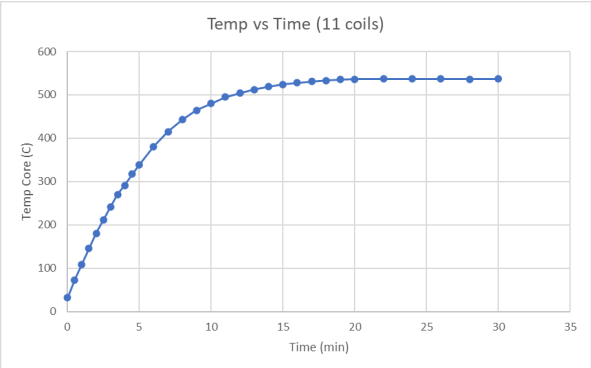
Time (min)	Temp Core (C)
0	25
0.5	43.6
1	72.4
1.5	102.4
2	130.3
2.5	158.2
3	183.4
3.5	208
4	231
4.5	252
5	270
6	303
7	332



8	355
9	374
10	391
11	405
12	415
13	427
14	436
15	445
16	451
17	456
18	461
19	464
20	466
22	470
24	472
26	473
28	474
30	475

11 coils

Time (min)	Temp Core (C)
0	32
0.5	73.1
1	108.4
1.5	145
2	180.2
2.5	211
3	241

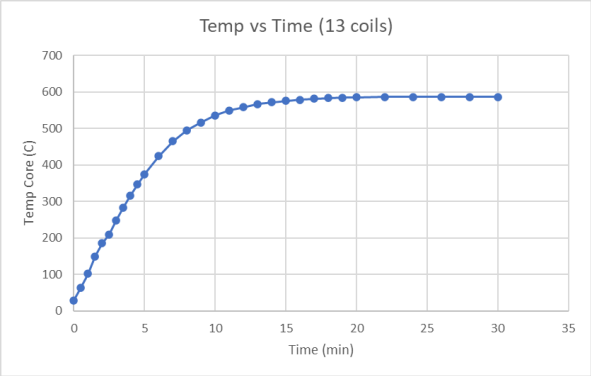


3.5	270
4	291
4.5	317
5	339
6	381
7	415
8	443
9	464
10	480
11	495
12	504
13	512
14	519
15	524
16	528
17	531
18	533
19	535
20	536
22	537
24	537
26	537
28	536
30	537

13 coils

Time (min)	Temp Core (C)
0	28.9

0.5	63.9
1	102
1.5	150
2	186
2.5	210
3	249
3.5	284
4	317
4.5	347
5	375
6	424
7	465
8	495
9	517
10	536
11	550
12	558
13	567
14	572
15	576
16	579
17	582
18	584
19	585
20	586
22	587
24	587
26	587
28	587
30	587



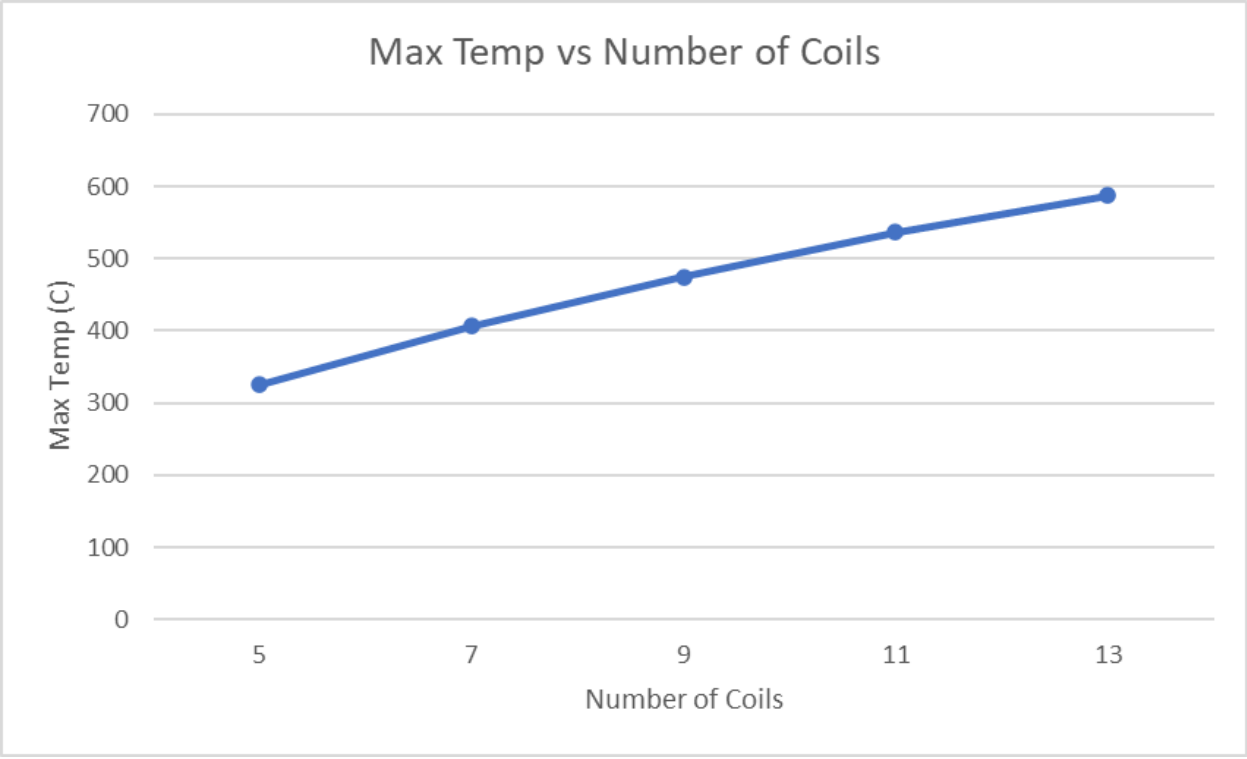
*Aluminum at 5amps and 13 coils reached 587 as well

Max Temps:

The maximum temperatures for SS tests in air are: 325°C for 5 coils, 406°C for 7 coils, 475°C for 9 coils, 537°C for 11 coils, and 587°C for 13 coils. Maximum temperature for SS test in vacuum is: 362°C for 5 coils

Max Temp vs #Coils

Coils	Max temp
5	325
7	406
9	475
11	537
13	587



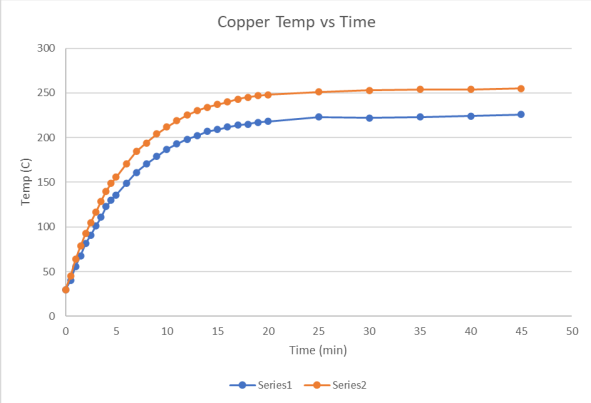
Approximate Equation for line:

$$T_m = 32.75C + 171.25$$

Copper Air Test @5amps

4 coils

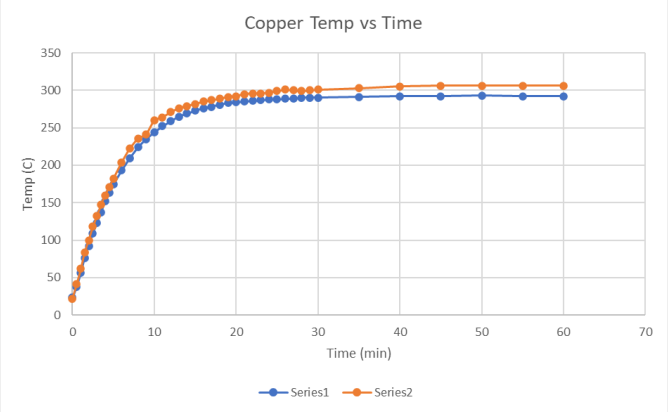
Time (min)	TOP (S1)(C)	BOTTOM (S2)(C)
0	30	30
0.5	40	45
1	56	64
1.5	67.8	78.6
2	81.5	92.5
2.5	90.6	104.5
3	101.4	117
3.5	110.9	128.8
4	122.9	140
4.5	129.7	148.8
5	135.8	155.7
6	148.9	170.6
7	161	184.5
8	170.5	194
9	179	204
10	186.9	212
11	192.9	219
12	198	225
13	202	230
14	207	234



15	209	237
16	212	240
17	214	243
18	215	245
19	217	247
20	218	248
25	223	251
30	222	253
35	223	254
40	224	254
45	226	255

5 coils

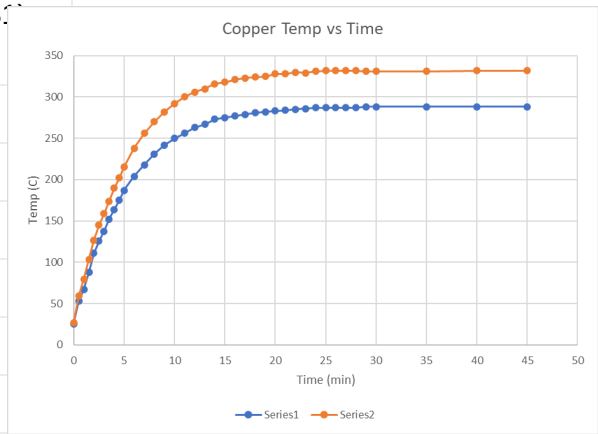
Time (min)	TOP (S1) (C)	BOTTO (C)
0	23.5	22.2
0.5	37.3	41.1
1	56.5	61.8
1.5	76	83.6
2	92.3	99.7
2.5	109	118
3	123.4	132.5
3.5	137.2	147.5
4	152	159.7
4.5	163.7	170.6
5	174.3	181.7
6	193.3	204
7	209	222
8	224	236



9	235	241
10	244	260
11	252	264
12	259	271
13	265	276
14	269	279
15	273	282
16	276	285
17	278	287
18	281	289
19	283	291
20	284	292
21	285	295
22	286	296
23	287	296
24	288	297
25	288	299
26	289	301
27	289	300
28	290	299
29	290	300
30	290	301
35	291	303
40	292	305
45	292	306
50	293	306
55	292	306
60	292	306

6 coils

Time (min)	TOP (S1) (C)	BOTTOM (S1) (C)
0	25.6	26.6
0.5	52.8	59
1	67.1	79.5
1.5	87.5	103.5
2	110.7	126.5
2.5	125.4	144.9
3	137.3	158.7
3.5	152.3	173.8
4	163.6	189.8
4.5	175	202
5	186.8	215
6	204	238
7	218	256
8	231	270
9	242	282
10	250	292
11	256	300
12	263	306
13	267	310
14	273	316
15	275	318
16	277	321
17	279	323
18	281	324
19	282	325
20	283	328



21	284	328
22	285	330
23	286	329
24	287	331
25	287	332
26	287	332
27	287	332
28	287	332
29	288	331
30	288	331
35	288	331
40	288	332
45	288	332

Vacuum Testing

Empty Baseline

8/9/23

2:44

3:04

4:10

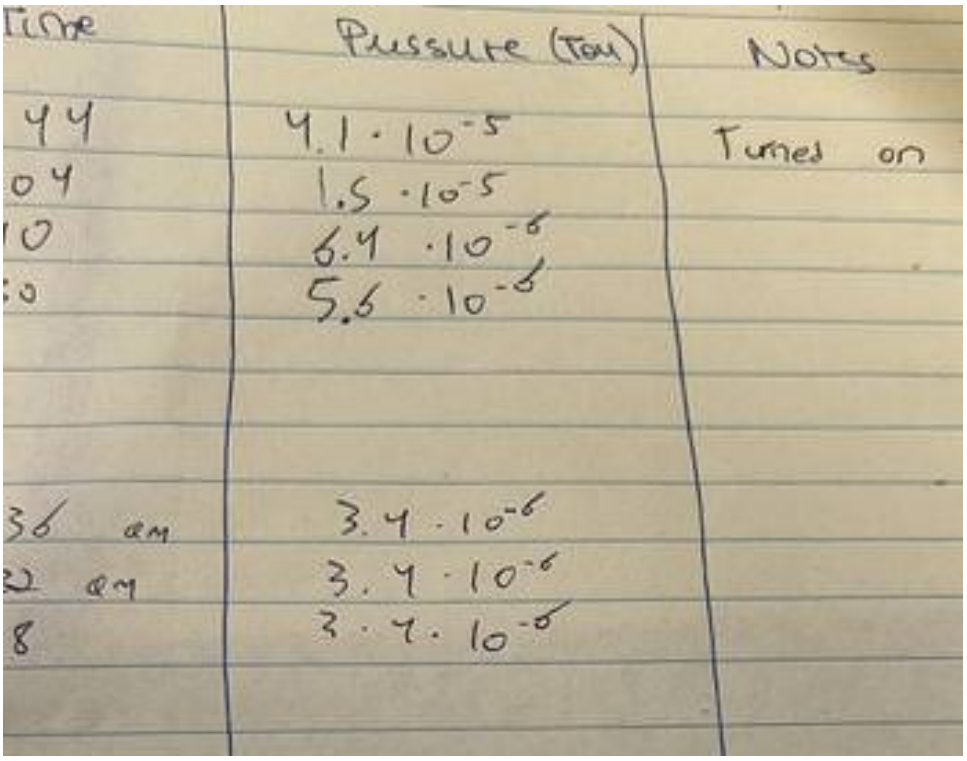
4:30

8/10/23

8:36

10:32

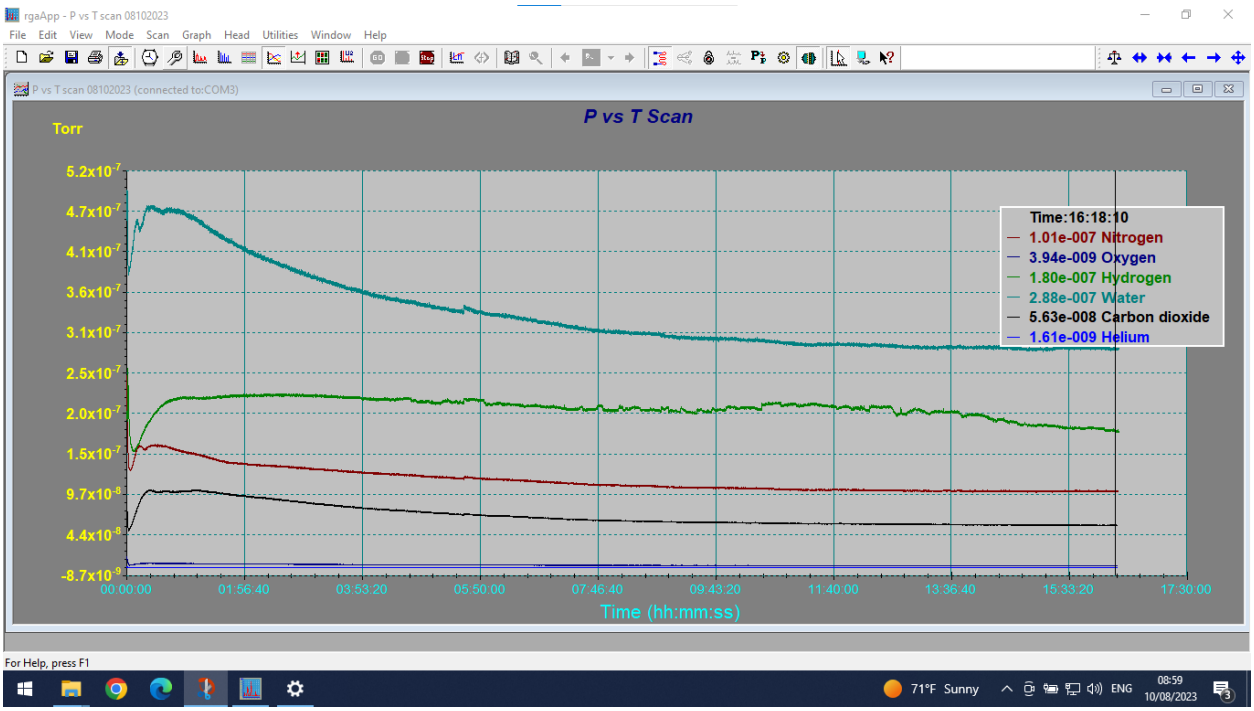
12:58



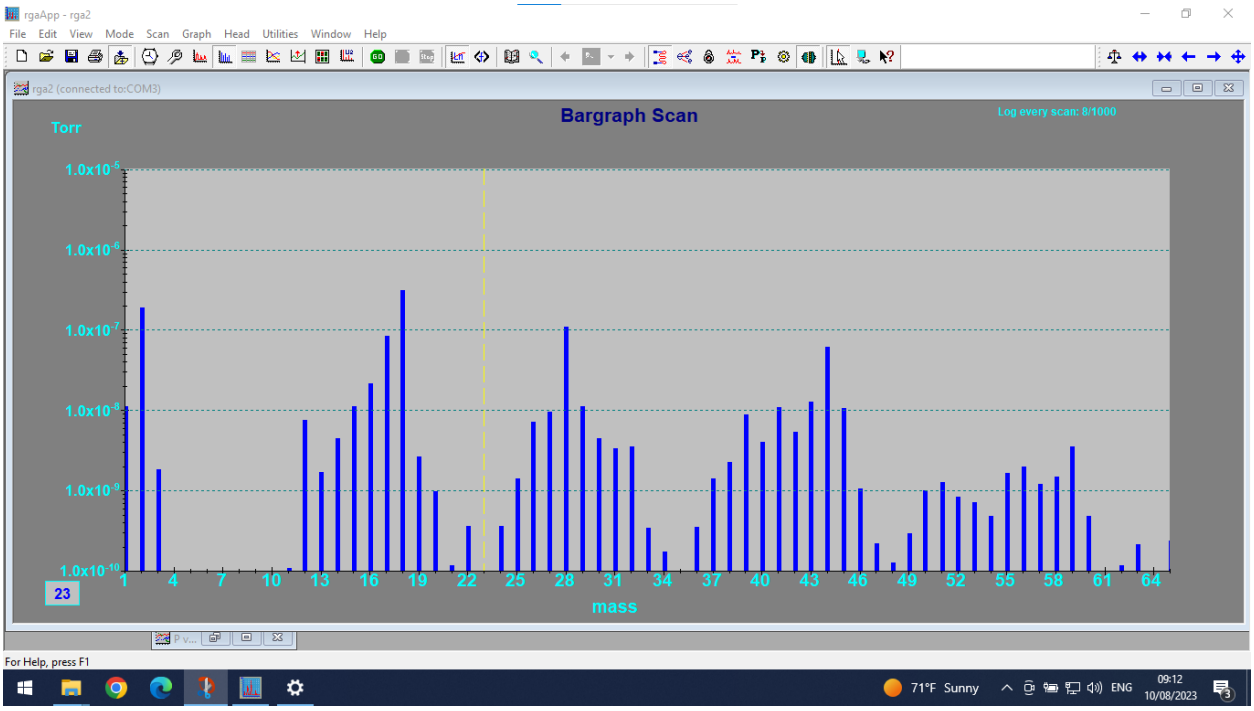
Time	Pressure (Torr)	Notes
44	$4.1 \cdot 10^{-5}$	Turned on
04	$1.5 \cdot 10^{-5}$	
10	$6.4 \cdot 10^{-6}$	
:0	$5.6 \cdot 10^{-6}$	
36 am	$3.4 \cdot 10^{-6}$	
2 am	$3.4 \cdot 10^{-6}$	
8	$3.4 \cdot 10^{-6}$	

Started at 1131 torr and over ~24 hour dropped to 3.4E-6

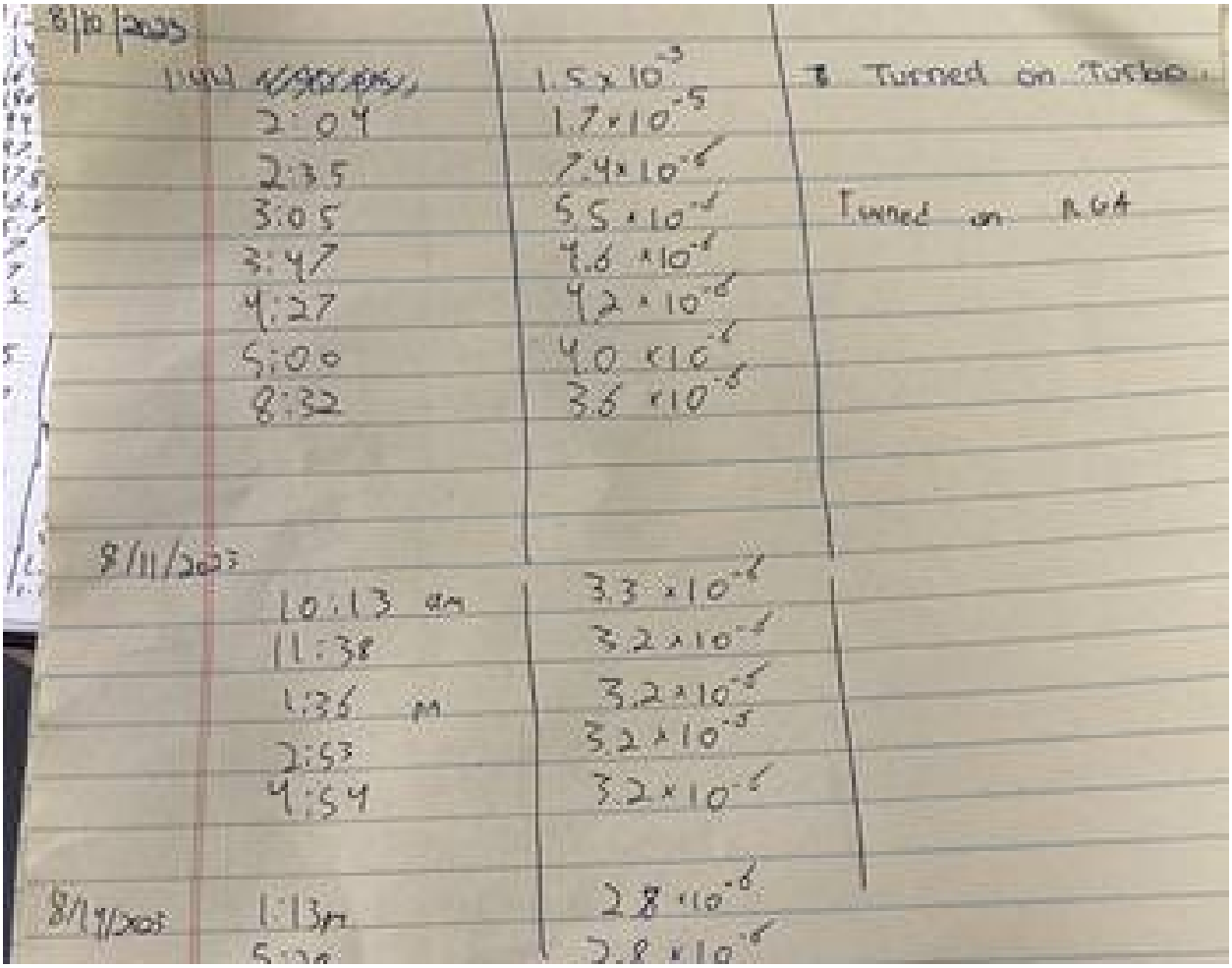
Final Partial Pressures:



Sum partial pressures 6.31E-7

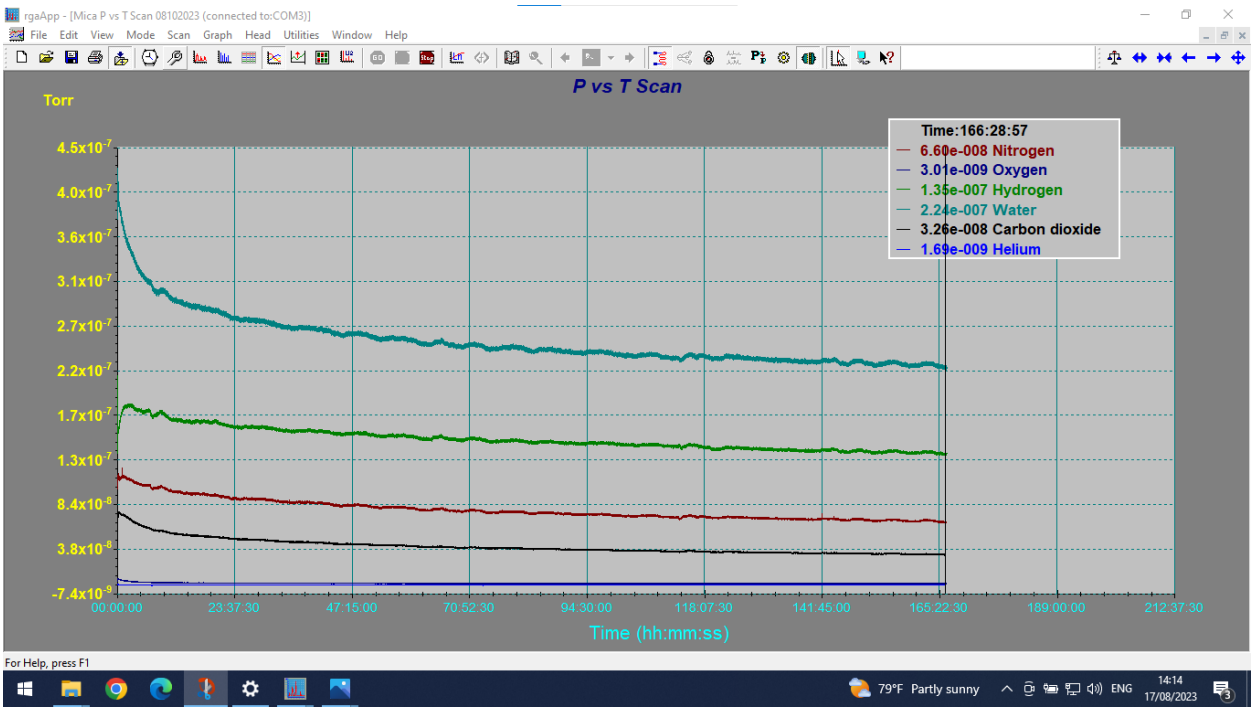


Mica Baseline

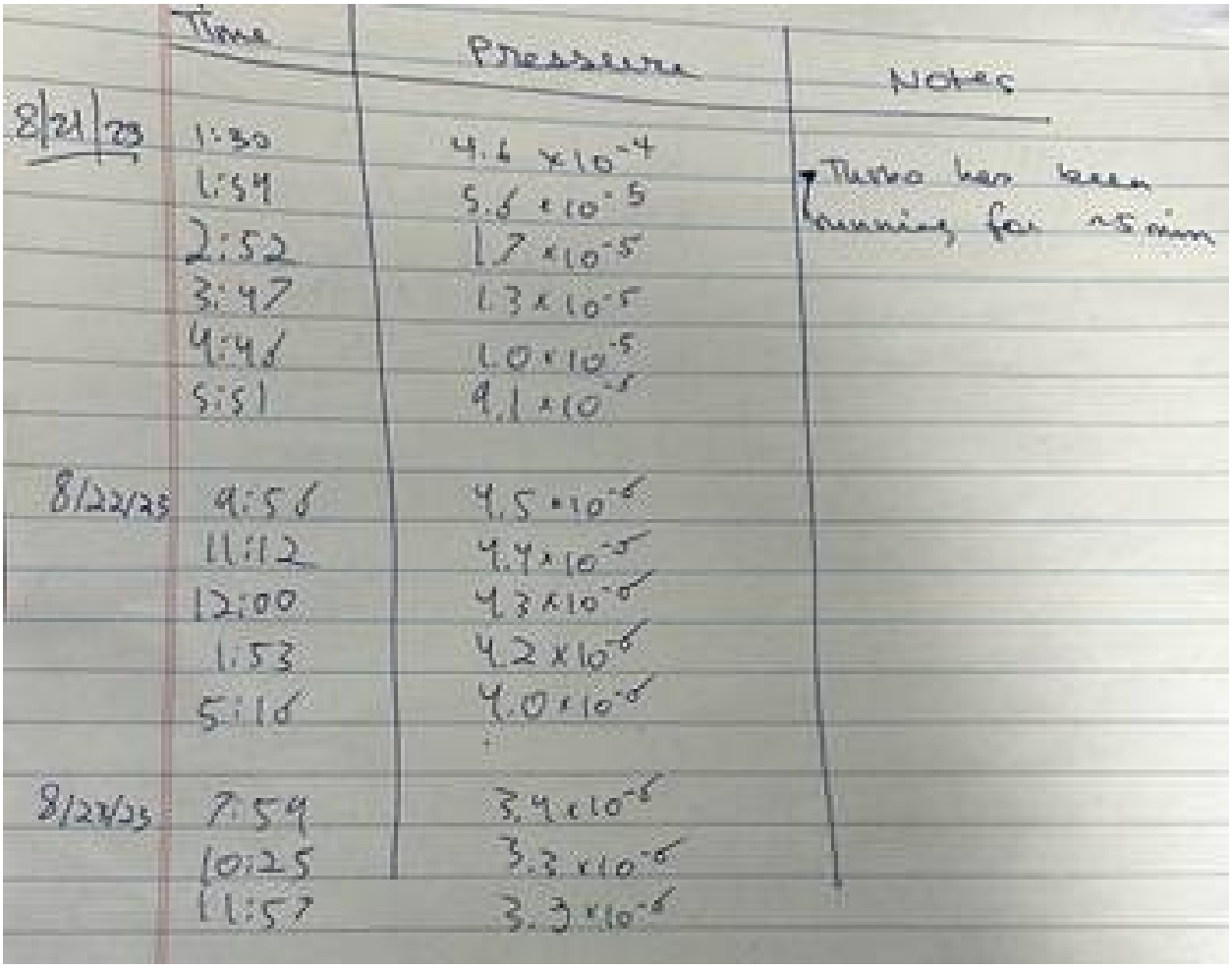


Started at 1131 torr, after 24 hours dropped to 3.2E-6, after ~124 hours dropped to 2.5E-6

Final Partial Pressures:



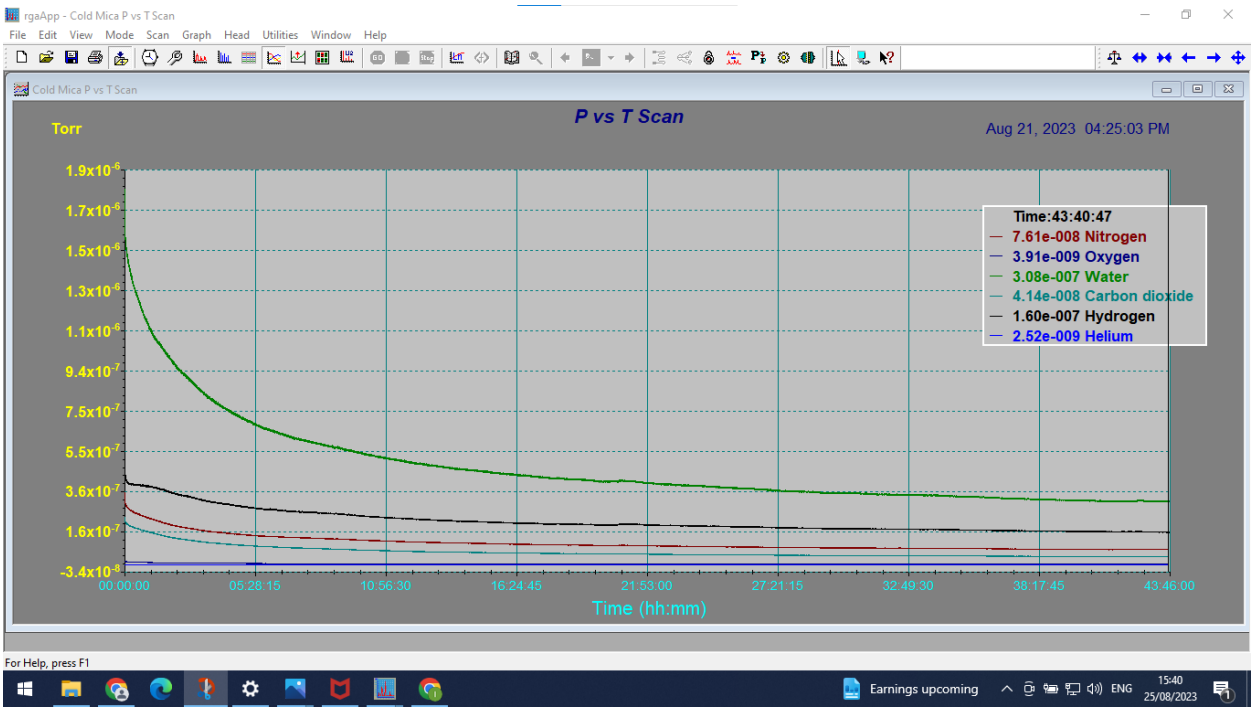
Mica + SS Reservoir Baseline

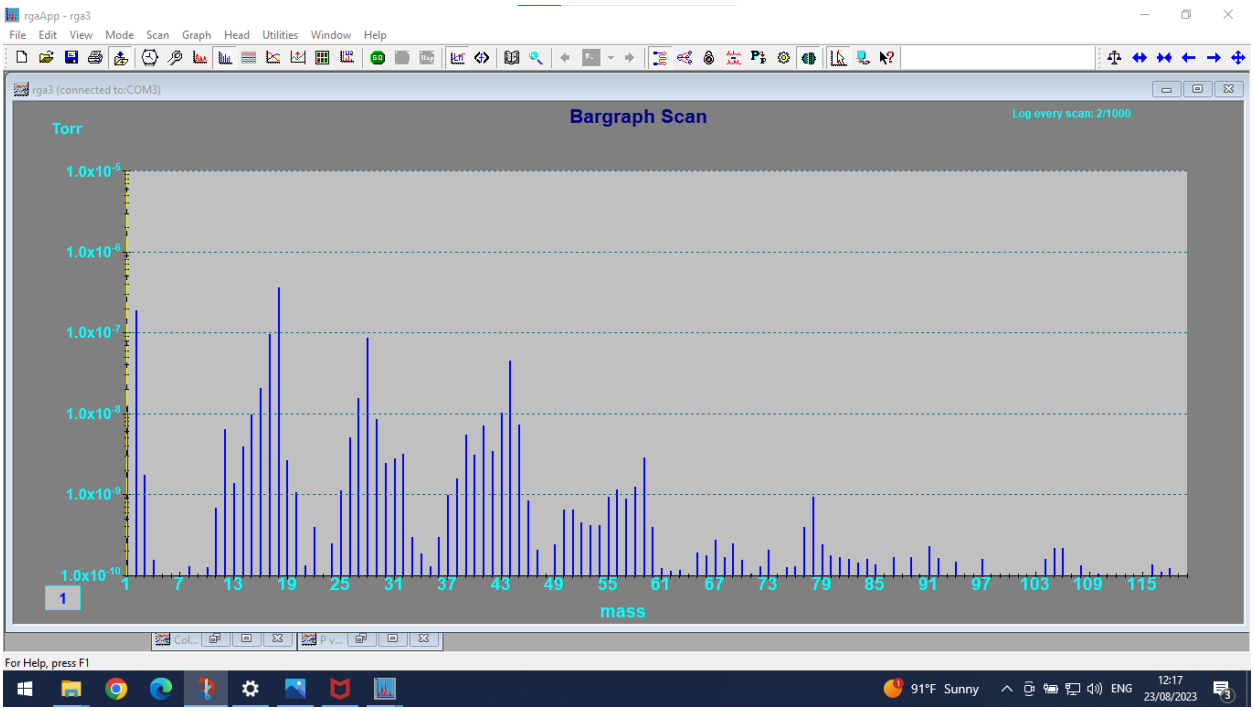


	Time	Pressure	Notes
8/21/23	1:30	4.4×10^{-4}	Turbo has been running for ~5 min
	1:54	5.6×10^{-5}	
	2:52	1.7×10^{-5}	
	3:47	1.3×10^{-5}	
	4:46	1.0×10^{-5}	
	5:51	9.1×10^{-6}	
8/22/23	9:56	4.5×10^{-6}	
	11:12	4.4×10^{-6}	
	12:00	4.3×10^{-6}	
	1:53	4.2×10^{-6}	
	5:16	4.0×10^{-6}	
8/23/23	7:59	3.9×10^{-6}	
	10:25	3.3×10^{-6}	
	11:57	3.3×10^{-6}	

Started at ~1130 torr, after ~24 hours dropped to 4.3E-6 torr, after ~70 hours dropped to 3.3E-6

Final Partial Pressures/Partial Pressure Before Heating:

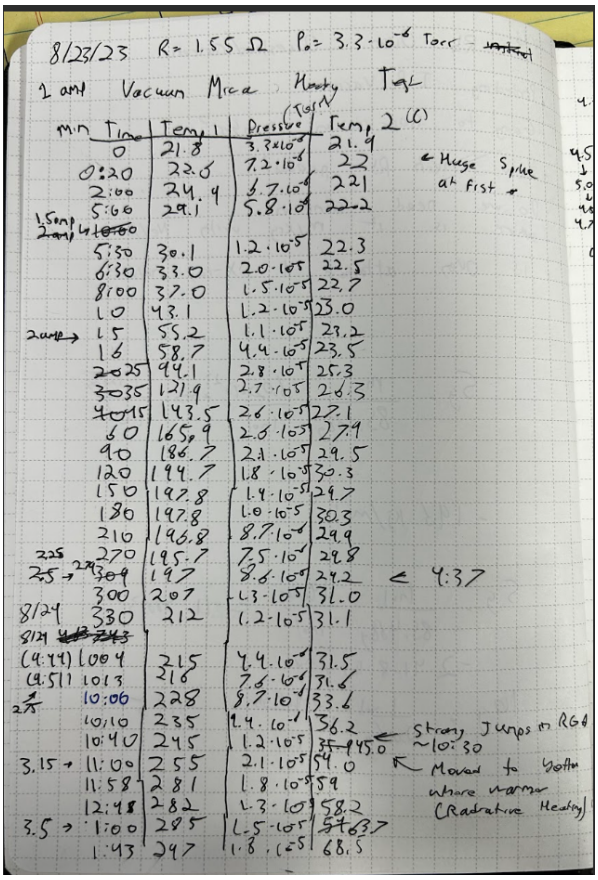




Total pressure: 3.3E-6

Heated Mica + SS Reservoir Test

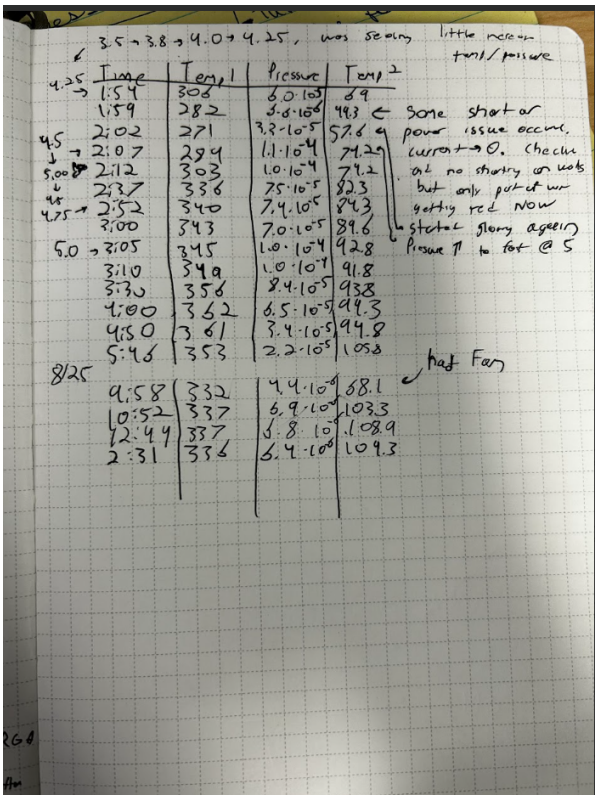
5 coils, R = 1.55 oms, initial pressure before heating P = 3.3E-6 torr



8/23/23 R = 1.55 Ω P_0 = 3.3·10⁻⁶ Torr

1 amp Vacuum Mica Heater Test

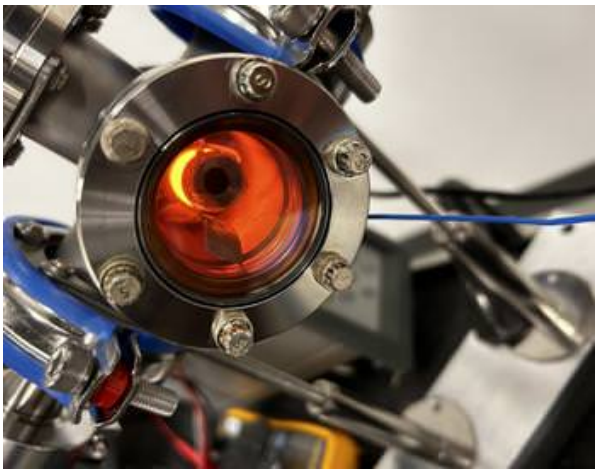
min	Time	Temp 1	Pressure (Torr)	Temp 2 (C)	
	0	21.8	3.3x10⁻⁶	21.9	
	0:20	22.0	7.2·10⁻⁶	22	← Huge Spike at first
	2:00	24.4	6.7·10⁻⁶	22.1	
	5:00	29.1	5.8·10⁻⁵	22.2	
1.5amp 2amp	4 ~~5:00~~				
	5:50	30.1	1.2·10⁻⁵	22.3	
	6:30	33.0	2.0·10⁻⁵	22.5	
	8:00	37.0	1.5·10⁻⁵	22.7	
	10	43.1	1.2·10⁻⁵	23.0	
2amp →	15	55.2	1.1·10⁻⁵	23.2	
	16	58.7	4.4·10⁻⁵	23.5	
	~~20~~ 25	94.1	2.8·10⁻⁵	25.3	
	~~30~~ 35	121.9	2.7·10⁻⁵	26.3	
	~~40~~ 45	143.5	2.6·10⁻⁵	27.1	
	60	165.9	2.6·10⁻⁵	27.9	
	90	186.7	2.1·10⁻⁵	29.5	
	120	194.7	1.8·10⁻⁵	30.3	
	150	192.8	1.4·10⁻⁵	29.7	
	180	197.8	1.0·10⁻⁵	30.3	
	210	196.8	8.7·10⁻⁶	29.9	
225	270	195.7	7.5·10⁻⁶	29.8	
25 → 25	~~309~~	197	8.6·10⁻⁶	29.2	← 4:37
	300	207	1.3·10⁻⁵	31.0	
8/24	330	212	1.2·10⁻⁵	31.1	
8/24	~~4:43~~ ~~2:43~~				
(9:44)	1004	215	4.4·10⁻⁶	31.5	
(9:51)	1013	216	7.6·10⁻⁶	31.6	
25	10:06	228	8.7·10⁻⁶	33.6	
	10:10	235	9.4·10⁻⁶	36.2	← Strong Jumps to RGA ~10:30
	10:40	245	1.2·10⁻⁵	~~35~~ 45.0	
3.15 →	11:00	255	2.1·10⁻⁵	54.0	↖ Moved to hotter where warmer (Radiative Heating)
	11:58	281	1.8·10⁻⁵	59	
	12:48	282	1.3·10⁻⁵	58.2	
3.5 →	1:00	295	1.5·10⁻⁵	~~57~~ 63.7	
	1:43	297	1.8·10⁻⁵	68.5	



3.5 → 3.8 → 4.0 → 4.25, was seeing little increase in temp/pressure

	Time	Temp 1	Pressure	Temp 2	
4.25 →	1:54	308	6.0·10⁻⁵	89	
	1:59	282	3.8·10⁻⁵	99.3	← Some start or power issue occurs, current → 0. Checks out no shorts or vents but only part of wire getting red Now
4.5 →	2:02	271	3.3·10⁻⁵	57.6	
5.00 →	2:07	294	1.1·10⁻⁴	74.29	
	2:12	303	1.0·10⁻⁴	79.2	
4.5 →	2:37	336	75·10⁻⁵	82.3	
4.75 →	2:52	340	7.4·10⁻⁵	84.3	
	3:00	343	7.0·10⁻⁵	89.6	↳ started glowing again
5.0 →	3:05	345	1.0·10⁻⁴	92.8	Pressure ↑ to for @ 5
	3:10	349	1.0·10⁻⁴	91.8	
	3:30	356	8.4·10⁻⁵	93.8	
	4:00	362	6.5·10⁻⁵	99.3	
	4:50	361	3.4·10⁻⁵	94.8	
	5:46	353	2.2·10⁻⁵	105.8	
8/25					↙ had Fan
	9:58	332	4.4·10⁻⁶	68.1	
	10:52	337	6.9·10⁻⁶	103.3	
	12:44	337	6.8·10⁻⁶	108.9	
	2:31	336	6.4·10⁻⁶	109.3	

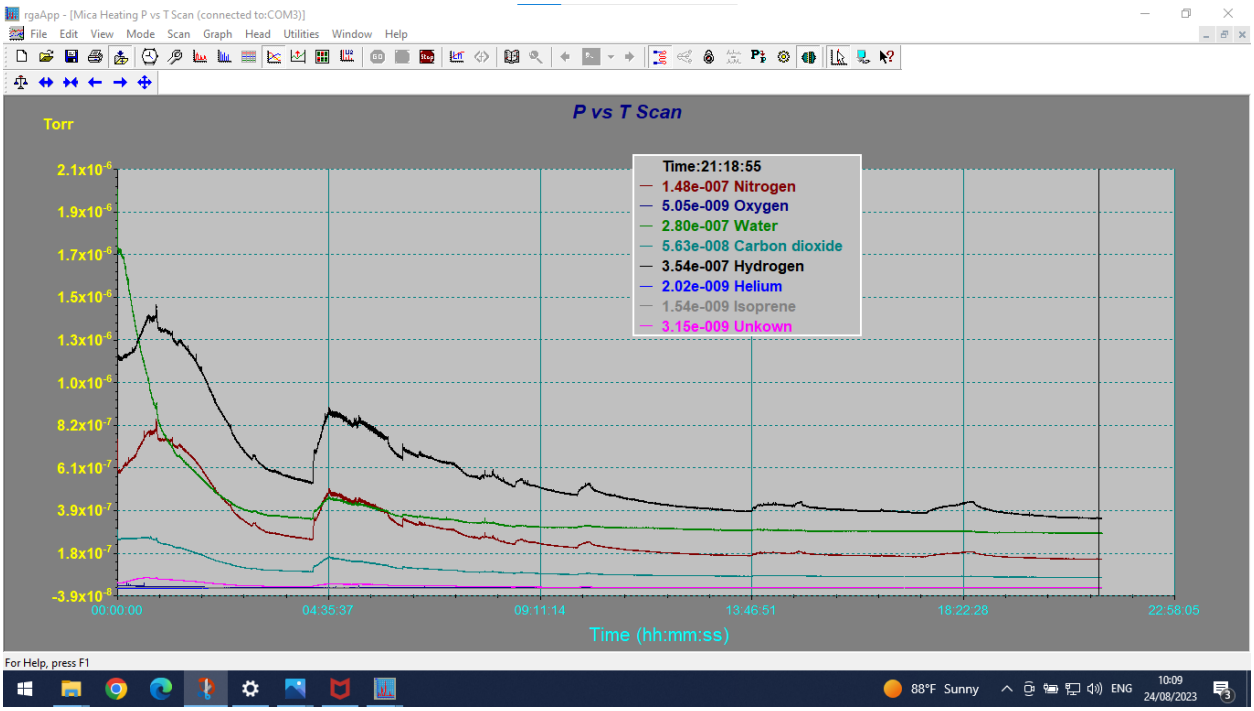
Slowly increased to 5 amps till reached max temp of 362 C. Outside also reached 109.3 C possibly due to part of wire touching mica touching the wall and radiative heating.





Started glowing around 2.75 amps or 215 C.

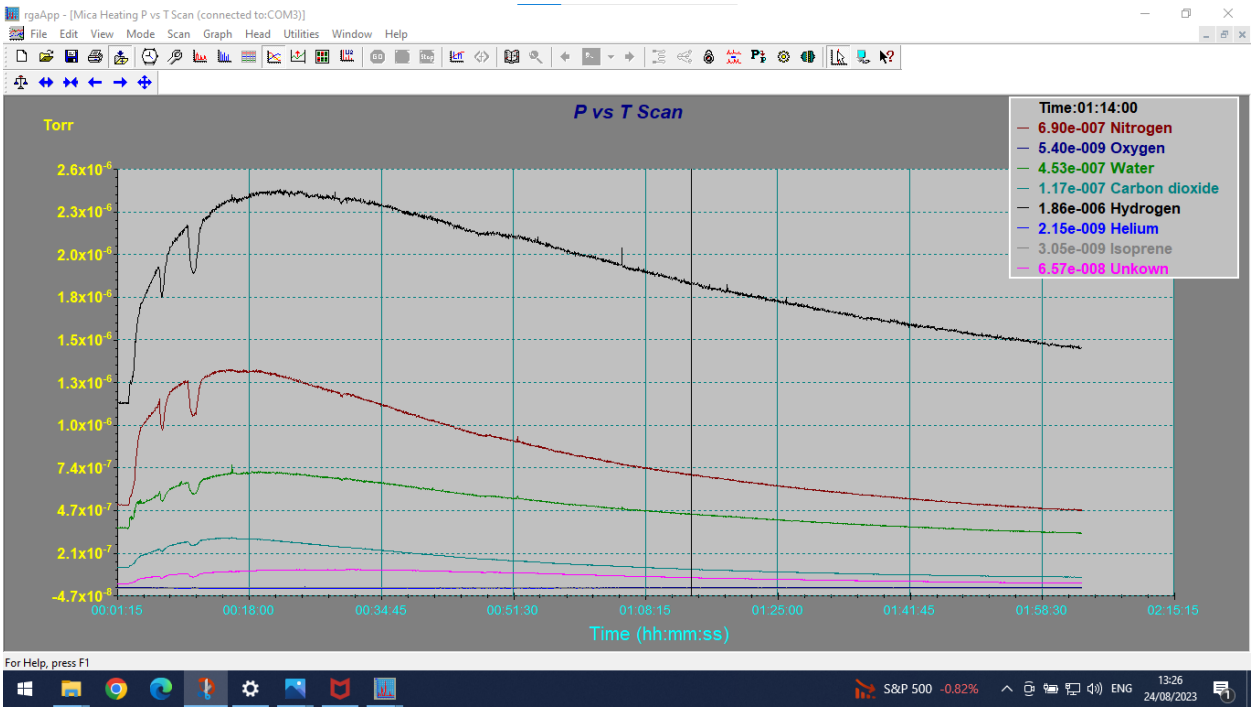
Partial Pressure Vs Time:



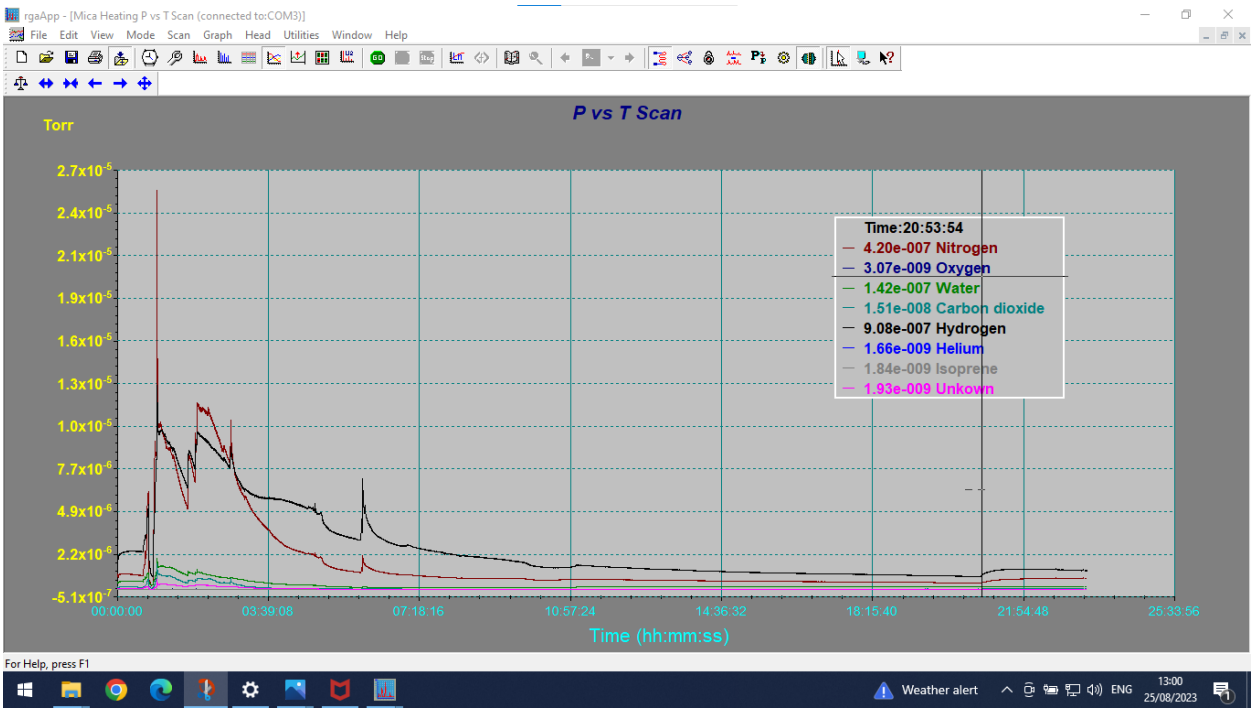
^ started 12:00 pm 8/23/23 at 2 amps. 4:00 pm jumped to 2.25 amps, sat over night



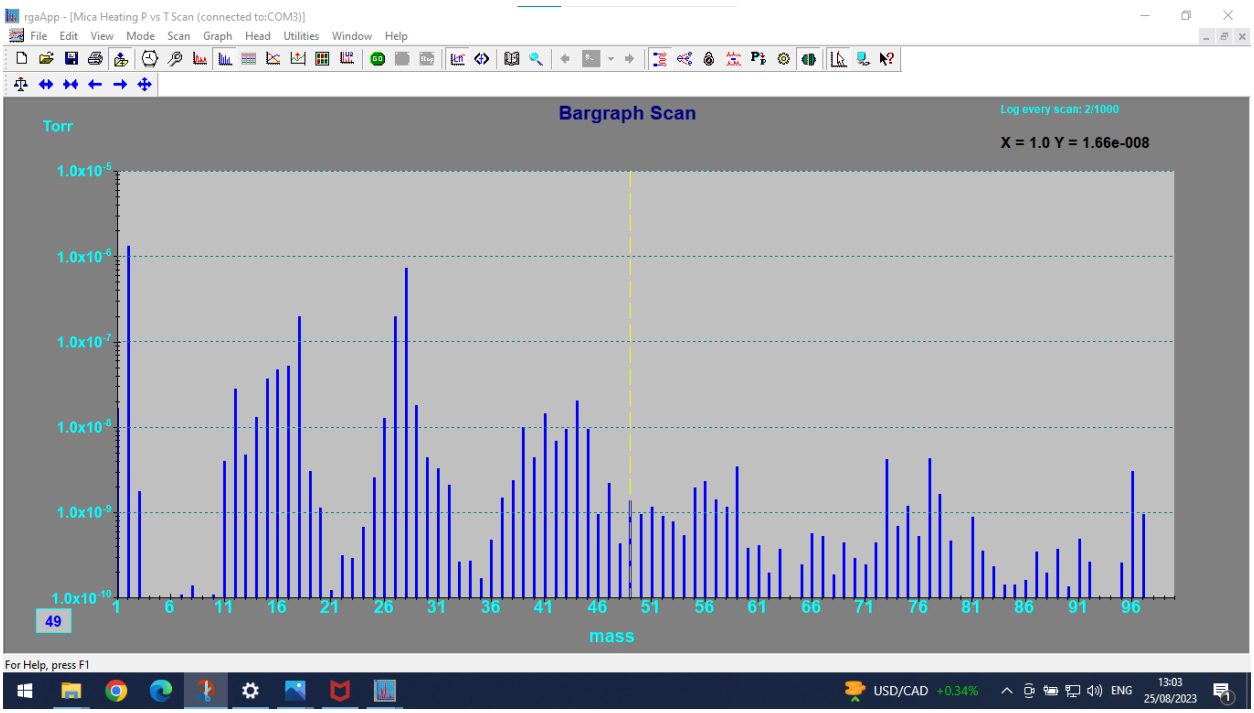
^started 10:00 am 8/24/23 at 2.75 amps, 11:00 am jumped to 3.15 amps



^started 11:20 am 8/24/23 at 3.15 amps

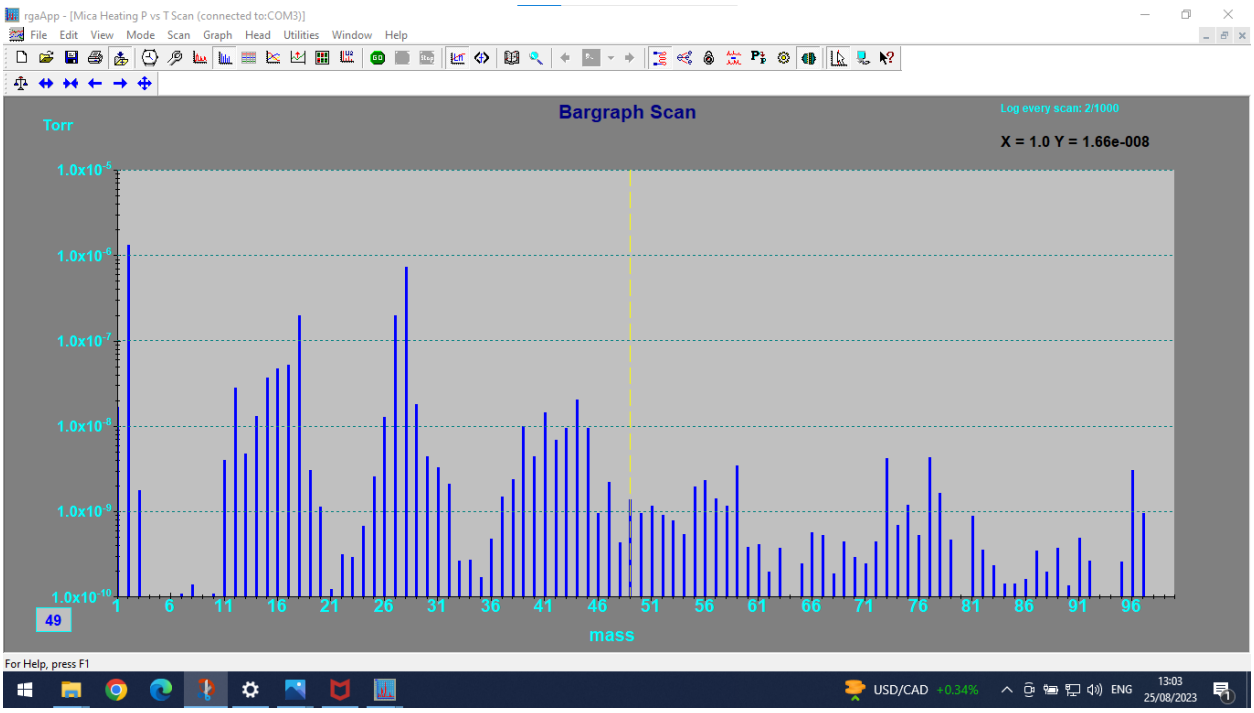


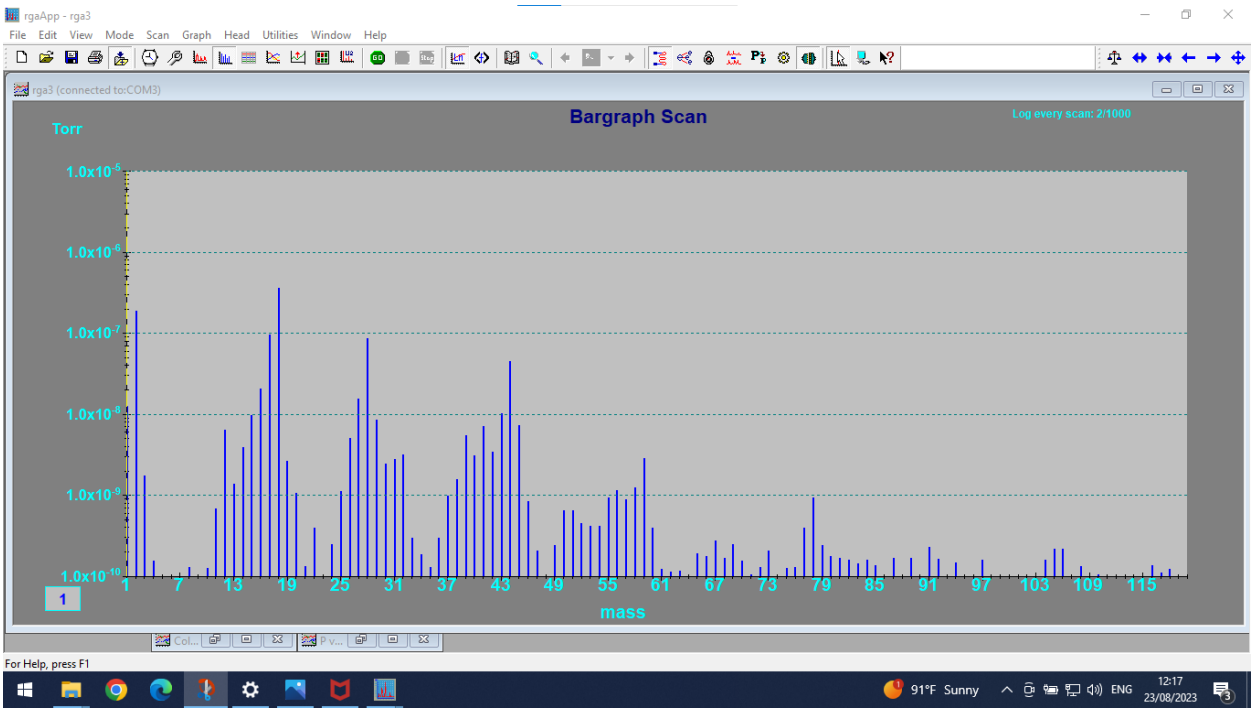
^started 1:30 pm 8/24/23. 1:50 pm jumped to 4.25 amps, 2:10 pm jumped to 4.5 (accidently went to 5 for a second correlating to huge spike in nitrogen and helium seen before going back down to 4.5), 2:52 pm jumped to 4.75 amps, 3:05 jumped to 5.0 amps.



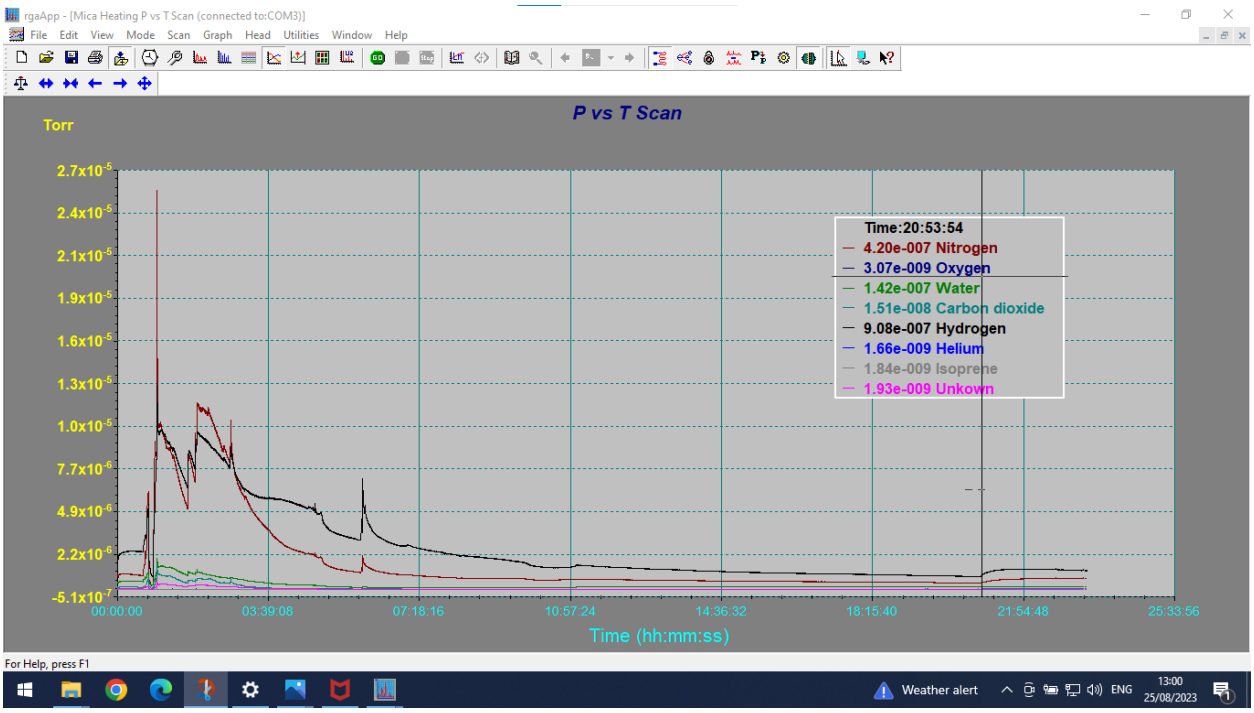
^Taken 12:40 pm 8/25/223. Total pressure 6.8E-6

Comparing Hot and Cold Pressures:

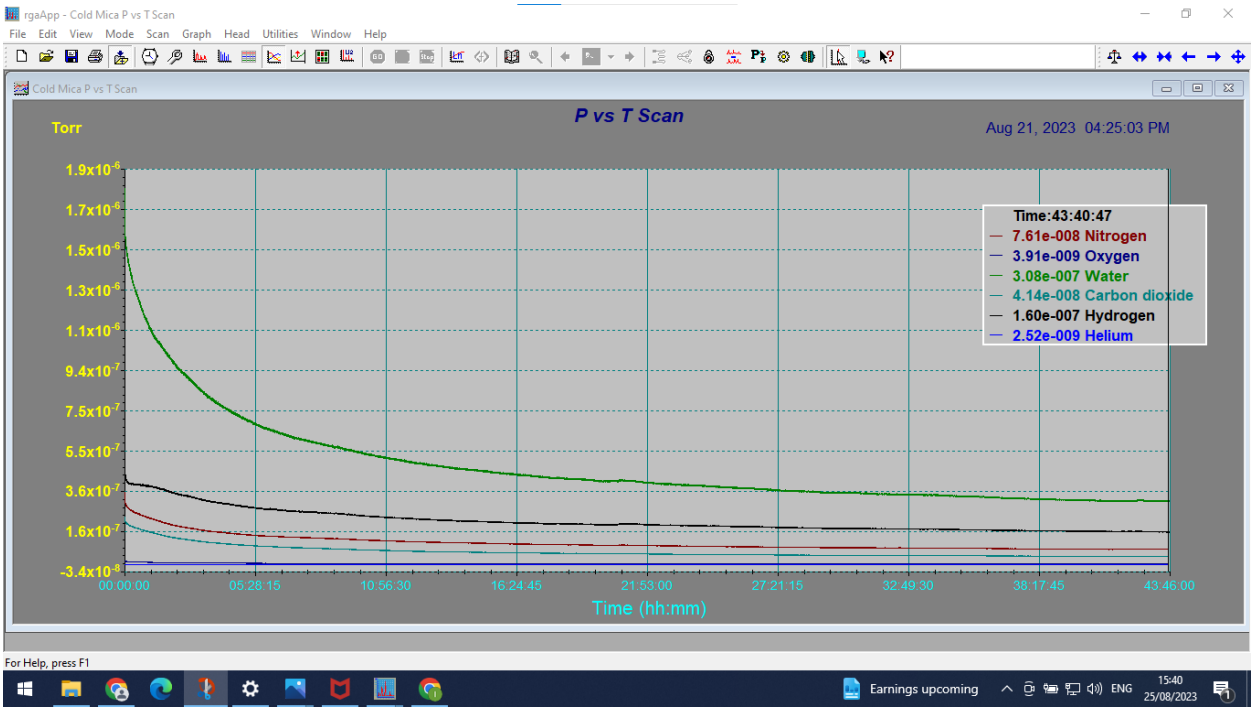




(Top is after heating, bottom is before heating) (Note: before heating has range 1-120, but after heating only 1-100) (Note: before heating total pressure was 3.3E-6, after heating 6.8E-6, more than double)



Sum of partial pressures 14.9E-7 (not including isoprene and unknown)



Sum of partial pressures: 5.92E-7 (consistent with empty cross: total pressure 3.4E-6, sum partial pressure 6.31E-7)

Difference in partial pressures of each gass in heat compared to cold test:

Nitrogen: 1.86x

Oxygen: 0.79x

Water: 0.46x

Carbon Dioxide: 0.36x

Hydrogen: 1.76x

Helium: 0.65x

Chemical Compositions:

Isoprene: C5H8

Mica:

CHEMICAL COMPOSITION OF MICA V1	
Silica (SiO_2)	45.57%
Alumina (Al_2O_3)	33.10%
Potassium Oxide (K_2O)	9.87%
Ferric Oxide (Fe_2O_3)	2.48%
Sodium Oxide (Na_2O)	0.62%
Titanium Oxide (TiO_2)	Traces
Calcium Oxide (CaO)	0.21%
Magnesia (MgO)	0.38%
Moisture at 100°C	0.25%
Phosphorus (P)	0.03%
Sulphur (S)	0.01%
Graphite Carbon (C)	0.44%
Loss on Ignition (H_2O)	2.74%